

Matthew B. Little

Earth and Atmospheric Systems Scientist
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PROFILE

Earth and Atmospheric Systems Scientist specializing in scientific data analysis, weather and climate modeling, and interdisciplinary collaboration across federal, academic, and research sectors. Recent experiences include analyzing Earth observing instrument datasets for intercalibration (NASA CLARREO Pathfinder Mission), developing and teaching meteorology curriculum (East Carolina University), and supporting CHIPS Metrology Program research (NIST).

PROFESSIONAL EXPERIENCE

IT Specialist (Remote); National Institute of Standards and Technology (NIST); Gaithersburg, MD

Jul 2024 – Oct 2025

Facilitated development, operation, and maintenance of data ingest capabilities for the CHIPS/METIS project in the Office of Information Systems Management (Awarded Noteworthy Recognition).

- Acted as Subject Matter Expert (SME) for technical services.
- Developed, tested, and deployed software solutions for infrastructure storage, directory, and data management systems.
- Provided IT liaison and program activity support services.

Adjunct Faculty Lecturer (Remote); East Carolina University (ECU); Greenville, NC

Jan 2024 – Dec 2024

Developed and taught undergraduate meteorology curriculum to 100 students each semester and provided individualized supplemental guidance to support flexible and accessible student learning.

- Designed and delivered engaging asynchronous learning modules using Canvas.
- Enhanced student success with customized academic support on request and for struggling students.
- Assessed student participation, essays, and exams.

Scientific Analyst (Remote); Science Systems and Applications, Inc. (SSAI), ADNET Systems; Lanham, MD

May 2022 – May 2024

Conducted Earth observing instrument and solar spectrometer data analysis to support the NASA CLARREO Pathfinder mission at NASA Langley Research Center.

- Developed Python scripts for the analysis and visualization of multi-instrument Earth observation datasets, reducing intercalibration methodology uncertainty below the 0.3% mission objective threshold.
- Isolated and quantified inter-instrument observation differences to minimize reflectance measurement uncertainty.
- Coauthored and presented scientific analysis at international conferences and interdisciplinary team meetings.

Graduate Research/Teaching Assistant (Hybrid); East Carolina University (ECU); Greenville, NC

Aug 2018 – May 2022

Conducted climate dynamics research for an NSF-funded project, estimating simulated seasonal precipitation changes in a warming climate.

- Configured, ran, and validated computer climate simulations, managing over 50 TB of model output.
- Developed data and model output processing and visualization scripts using Python, IDL, Matlab, and bash.
- Coauthored and presented research at international conferences.
- Supported Geography Department instructional, recruitment, and administrative functions by guest lecturing and grading assignments for 100+ students, meeting and guiding prospective graduate students, and updating the departmental website.

Texas Science Olympiad Meteorology Event Director (Hybrid); Texas A&M University; College Station, TX
Oct 2013 – Mar 2018

Designed and administered the annual Texas Science Olympiad Meteorology event exam for 200 participants, managed 3 event volunteers, and coordinated exam evaluation processes.

- Authored the Meteorology event exam, aligning content with national guidelines.
- Designed and led exam preparedness workshops attended by over 100 students, coaches, and supporters, while also creating publicly available video presentations for broader accessibility and guidance.
- Created and managed an educational website to provide participants with relevant academic resources.
- Reviewed and judged student submissions for affiliated science competitions.

Hydrologist (Hybrid); United States Forest Service (USFS); St. Ignace, MI
Jan 2017 – Feb 2018

Conducted hydrologic and soil inventories and assessments to inform land management planning, water resource improvement projects, and forest and wetland sustainability practices while identifying water resource management goals.

- Documented NEPA and CWA compliance in impacted wildland and plantation areas.
- Collaborated with interdisciplinary teams to document and improve forest and wetland hydrology.
- Developed ArcMap hydrologic and soil feature data visualizations to support project reports.

Graduate Assistant Researcher (Hybrid); Texas A&M University; College Station, TX
Jul 2012 – Sep 2013

Oversaw HPC ocean model simulations, modified model code, and automated model output processing.

- Collaborated with team to develop and refine research methodologies.
- Delivered literature reviews and progress updates during research team meetings.

Student Researcher (Hybrid); San Jose State University (SJSU); San Jose, CA
Sep 2010 – Dec 2011

(1) Developed a wind climatology utilizing weather model output analysis and visualizations to identify optimal urban wind turbine locations across the California Bay Area. (2) Developed a carbon emissions model linking university commuter students' automobile carbon emissions to university mixed-mode course delivery offerings.

- Developed data ingest, modeling, and output processing pipelines using IDL, Matlab, Bash, and Fortran.
- Proposed means to cut parking-related expenses by \$1 million/year, freeing 10 acres of land for SJSU.
- Coauthored scientific documents and presented results and financial impacts to university leadership.

Intern, Earth Data Analysis; Science Systems and Applications, Inc. (SSAI); Mountain View, CA
Jun 2011 – Aug 2011

Developed climate model output analysis to estimate regional atmospheric and environmental effects of climate change across California, supporting the Climate Adaptation Science Investigation Team at NASA Ames Research Center (Recipient of the NASA Group Achievement Award).

- Developed data organization, analysis, and visualization scripts using IDL and Matlab.
- Created ArcMap visualizations of projected environmental and atmospheric changes in future climate scenarios to support climate adaptation planning and protect high-value assets.
- Coauthored and presented scientific reports at international conferences, NASA events, and meetings.

Intern, Statistical Analysis; California State University, Long Beach (CSULB); Long Beach, CA
Jun 2009 – Aug 2009

Performed and presented statistical analysis of university restaurant consumer purchasing patterns.

- Developed methodology and scripts to clean, structure, and transform datasets.
- Performed statistical analysis using SPSS statistical software.
- Presented research results to multidisciplinary audiences.

Mathematics Tutor – Algebra and Calculus (Hybrid); Cypress Community College; Cypress, CA
Jan 2008 – Jun 2009

Facilitated structured group tutoring sessions and delivered tailored one-on-one support to resolve individual challenges, achieving a 10% average improvement in student rankings.

- Developed lesson plans and coordinated group activities.
- Supported students on an as-needed basis in a study hall environment.
- Presented mathematics teaching methodologies to education professionals at regional conferences.

EDUCATION

East Carolina University

Master of Science (M.S.) in Geography (2022)

Focus Area in Atmospheric Science

GPA: 4.00 / 4.00

Thesis: *WRF Simulations of the 2009 Southeastern United States Convective Season Onset in Current and Future Climate Scenarios*

San Jose State University

Bachelor of Science (B.S.) in Meteorology (2012)

Minor in Mathematics

GPA: 3.76 / 4.00

Thesis: *Modeling the Relationship Between Carbon Dioxide Emissions and Hybrid Course Availability at San Jose State University*

Austin Community College

Certificate in Computer Information Technology (2016)

GPA: 4.00 / 4.00

HIGHLIGHTED SKILLS

ArcMap (GIS); Bash; Confluence; Fortran; Git; GrADS; IDL; Interdisciplinary collaboration; Linux; Matlab; Python; Remote sensing; Scientific analysis, presentations, reports, and writing; Weather and climate modeling.

TECHNICAL SKILLS

Adobe Photoshop; CSS; High-performance computing; HTML; JavaScript; JupyterLab; Lucidchart; NCL; NOAA Weather and Climate Toolkit; Oracle; Panoply; PHP; SPSS; SQL; VS Code; Weather Research and Forecasting (WRF) modeling.

GENERAL SKILLS

Atmospheric science; Computer programming; Critical thinking; Data analysis; Mathematics; Mathematics education; Meteorology; Microsoft 365; Microsoft Office; Microsoft Windows; Physical oceanography; Science education; Science outreach; Tutoring; University lecturing.

PUBLICATIONS AND PRESENTATIONS

- Little, M.,** R. Bhatt, Y. Shea, 2023: Climate Absolute Radiance and Refractivity Observatory (CLARREO) Pathfinder Intercalibration Data Analysis Strategy. *2023 AGU Fall Meeting*, San Francisco, CA. 11 – 15 December 2023.
(Poster)
- Bhatt, R., Y. Shea, W. Wu, Q. Yang, D. Goldin, W. Sun, **M. Little**, X. Liu, N. Smith, C. Lukashin, 2023: CLARREO Pathfinder as a SI-traceable reference for satellite intercalibration. *SPIE Conference Proceedings 12685, Earth Observing Systems XXVIII*, San Diego, CA. 18 – 22 August 2023.
(Proceedings Paper)
- Little, M.,** 2022: WRF Simulations of the 2009 Southeastern United States Convective Season Onset in Current and Future Climate Scenarios. M.S. thesis, Dept. of Geography, Planning and Environment, East Carolina University.
(Thesis)
- Little, M.,** R. Ferreira, T. Rickenbach, 2020: WRF Simulations of the 2009 Southeastern United States Convective Season Onset in a Future Climate Scenario. *2020 AMS Annual Meeting*, Boston, MA. 11 – 16 January 2020.
(Poster)
- Little, M.,** E. Cordero, 2014: Modeling the Relationship Between Transportation–Related Carbon Dioxide Emissions and Hybrid–Online Courses at a Large Urban University. *International Journal of Sustainability in Higher Education*, 15, 3.
(Paper)
- Pitts, K., **M. Little**, M. Lowenstein, L. Iraci, C. Milesi, C. Schmidt, J.W. Skiles, 2012: Climate Adaptation Science Investigation (CASI) At NASA Ames Research Center: Using the Terrestrial Observation and Prediction System (TOPS) to Analyze Impacts of Climate Change on California Ecosystems. *2012 ASPRS Conference Proceedings*, Sacramento, CA. 1 – 5 March 2012.
(Proceedings Paper)
- Little, M.,** K. Pitts, M. Lowenstein, L. Iraci, C. Milesi, C. Schmidt, J.W. Skiles, 2011: Using the Terrestrial Observation and Prediction System (TOPS) to Analyze Impacts of Climate Change on California Ecosystems. Climate Change Science for Communities and Institutions, *2011 AGU Fall Meeting*, San Francisco, CA. 5 – 9 December 2011.
(Poster)
- Little, M.,** E. Cordero, 2011: Online Versus Face–to–Face Courses: An Analysis of Carbon Emissions. *College of Science Research Day*. Department of Meteorology, San Jose State University. San Jose, CA. 13 May 2011.
(Poster)
- Little, M.,** 2010: What is... the Lorenz Attractor? Department of Mathematics, San Jose State University. San Jose, CA. 15 November 2010.
(Presentation)